Indices, Surds and Polynomials

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Indices, Surds & Polynomials Practice Exercises**

At 14:00, a coastguard radar station detects two vessels, A and B. Relative to the radar station, vessel A has position vector $(-4\mathbf{i}+6\mathbf{j})$ km and moves with a constant velocity of $(8\mathbf{i}+5\mathbf{j})$ km h^{-1}. Vessel B has position vector $(20\mathbf{i}-4\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i}+10\mathbf{j})$ km h^{-1}, where u is a constant. Given that vessel B intercepts vessel A, find the value of u.

Let t be the time in hours after 14:00 when the interception occurs. The position vector of vessel A at time t is: $\mathbf{r}_A(t)=(-4+8t)\mathbf{i}+(6+5t)\mathbf{j}$ The position vector of vessel B at time t is: $\mathbf{r}_B(t)=(20+ut)\mathbf{i}+(-4+10t)\mathbf{j}$ For interception to occur, $\mathbf{r}_A(t)=\mathbf{r}_B(t)$. Equating the $\mathbf{j}$-components: $6+5t=-4+10t\Rightarrow 5t=10\Rightarrow t=2$ hours Equating the $\mathbf{i}$-components and substitu.

2 **[Foundational]** • **Target: Fractional Indices and Exponential Equations**

A sector of a circle of radius 12 cm has an arc length of 18 cm. Find the area of the sector.

The arc length s of a sector of radius r subtending an angle θ radians at the centre is given by $s=r\theta$. Substituting the given values gives $18=12\theta\Rightarrow\theta=1.5$ radians. The area A of the sector is given by $A=\frac{1}{2}r^2\theta=\frac{1}{2}(12)^2(1.5)=108$ cm^2 (or directly via $A=\frac{1}{2}rs=\frac{1}{2}\times 12\times 18=108$ cm^2). Omitting the factor of $\frac{1}{2}$ yields the incorrect value of 216 cm^2, while.

3 **[Foundational]** • **Target: Surds and Rationalising Denominators**

Given that ${}^nP_3=8\,{}^nC_2$, find the value of the integer n, where $n\geq 3$.

By definition of permutations and combinations: ${}^nP_3=\frac{n!}{(n-3)!}=n(n-1)(n-2)$ and ${}^nC_2=\frac{n!}{2!\,(n-2)!}=\frac{n(n-1)}{2}$. Substitute these expressions into the given equation: $n(n-1)(n-2)=8\left(\frac{n(n-1)}{2}\right)=4n(n-1)$. Since $n\geq 3$, $n(n-1)eq0$. Dividing both sides by $n(n-1)$ gives: $n-2=4\Rightarrow n=6$. Common error: Omitting the factor of $2!$ in the denominator of ${}^nC_2$ leads to $n-2=8\Rightarrow n=10$.

4 **[Foundational]** • **Target: Remainder and Factor Theorems**

The polynomial $p(x)=2x^3+ax^2-5x+6$ is divisible by $x-2$. Find the value of the constant a.

By the Factor Theorem, if $p(x)$ is divisible by $x-2$, then $p(2)=0$. Substitute $x=2$ into the polynomial expression: $p(2)=2(2)^3+a(2)^2-5(2)+6$ $p(2)=2(8)+4a-10+6=16+4a-10+6=4a+12$ Setting $p(2)=0$: $4a+12=0\Rightarrow 4a=-12\Rightarrow a=-3$ Common misconceptions: - Using the root $x=-2$ instead of $x=2$ leads to $2(-8)+4a-5(-2)+6=4a=0\Rightarrow a=0$. - A sign error when solving the linear equation $4a+12=0$ incorrectly gives $a=3$.

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5 [Foundational] • Target: The Binomial Theorem

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 5$ cm?

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 5$, $\frac{dV}{dt} = 4\pi(5)^2(0.5) = 2\pi(25) = 50\pi$ cm^3/s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

6 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Find all solutions of $4\cos^2 x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

Solving quadratic trig equation $4\cos^2 x - 3 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

7 [Foundational] • Target: Fractional Indices and Exponential Equations

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

Solving quadratic trig equation $2\cos^2 x + 3\sin x - 3 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Surds and Rationalising Denominators

Evaluate the definite integral: $\int_0^1 (6x+1)^3\,dx$.

Substitute $u = 6x + 1 \Rightarrow du = 6\,dx$. Integral: $\frac{1}{6}\left[\frac{u^4}{4}\right]_1^7 = \frac{(7)^4 - 1^4}{24} = \frac{2400}{24}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

9 [Intermediate] • Target: Remainder and Factor Theorems

A sector of a circle of radius 12 cm has a central angle of $\frac{\pi}{3}$ radians. Find the perimeter of the sector, giving your answer in exact form.

To find the perimeter of a sector, we add the arc length to the two bounding radii. 1. Calculate the arc length s: $s = r\theta = 12 \times \frac{\pi}{3} = 4\pi$ cm 2. Calculate the perimeter P: $P = 2r + s = 2(12) + 4\pi = 24 + 4\pi$ cm Analysis of incorrect choices:

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10 [Intermediate] • Target: The Binomial Theorem

In the expansion of $(1+\frac{x}{4})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n?

Coeff of x is $\frac{n}{4}$. Coeff of x^2 is $\frac{n(n-1)}{2(16)}$. Given $\frac{n(n-1)}{2\cdot 16}=2(\frac{n}{4})\Rightarrow\frac{n-1}{2\cdot 4}=2\Rightarrow n-1=4(4)=16\Rightarrow n=17$.

11 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The polynomial $P(x)$ is defined by $P(x)=(2+ax)^n+bx^4+cx-4$ where n is an integer greater than 2, and a, b, c are constants with $a>0$. In the binomial expansion of $(2+ax)^n$ in ascending powers of x: - the coefficient of x^2 is twice the coefficient of x, - the coefficient of x^3 is equal to the coefficient of x^2. Given that $(x+2)$ is a factor of $P(x)$, and that the remainder when $P(x)$ is divided by $(x+1)$ is 7, find the remainder when $P(x)$ is divided by $(x-1)$.

Final Answer: 1011. We solve the problem in three stages. **Stage 1: Determine a and n using the Binomial Theorem** The general term in the expansion of $(2+ax)^n$ is $\binom{n}{r}2^{n-r}(ax)^r$. - The term in x is $\binom{n}{1}2^{n-1}(ax)=n\cdot 2^{n-1}ax$. - The term in x^2 is $\binom{n}{2}2^{n-2}(ax)^2=\frac{n(n-1)}{2}2^{n-2}a^2x^2$. - The term in x^3 is $\binom{n}{3}2^{n-3}(ax)^3=\frac{n(n-1)(n-2)}{6}2^{n-3}a^3x^3$. Using the first condition (coefficient of x^2 is twice that.

12 [Intermediate] • Target: Fractional Indices and Exponential Equations

In the expansion of $(1+ax)^n$ in ascending powers of x, the coefficient of x^2 is 60 and the coefficient of x^3 is 160, where n is a positive integer and $a>0$. A cubic polynomial $P(x)$, in which the coefficient of x^3 is 1, has $(x-a)^2$ as a factor and leaves a remainder of n when divided by $(x-1)$. Find the remainder when $P(x)$ is divided by $(x+a)$.

Final Answer: 48. Step 1: Determine n and a using the Binomial Theorem. In the expansion of $(1+ax)^n$: Term in $x^2=\binom{n}{2}(ax)^2=\frac{n(n-1)}{2}a^2x^2=60x^2\Rightarrow n(n-1)a^2=120$ Term in $x^3=\binom{n}{3}(ax)^3=\frac{n(n-1)(n-2)}{6}a^3x^3=160x^3\Rightarrow n(n-1)(n-2)a^3=960$ Dividing the equation for x^3 by the equation for x^2.

13 [Intermediate] • Target: Surds and Rationalising Denominators

The polynomial $P(x)=2x^3+ax^2+bx-6$ has a factor of $x-2$ and leaves a remainder of -24 when divided by $x+1$, where a and b are constants. Find the remainder when $P(x)$ is divided by $x-3$.

Final Answer: 12. Step 1: Use the Factor Theorem for the factor $x-2$. Since $x-2$ is a factor, $P(2)=0$: $P(2)=2(2)^3+a(2)^2+b(2)-6=0$ $16+4a+2b-6=0\Rightarrow 4a+2b=-10\Rightarrow 2a+b=-5$ $---$ (1) Step 2: Use the Remainder Theorem for division by $x+1$. Since the remainder is -24, $P(-1)=-24$: $P(-1)=2(-1)^3+a(-1)^2+b(-1)-6=-24$ $-2+a-b-6=-24\Rightarrow a-b-8=-24\Rightarrow a-b=-16$ $---$ (2) Step 3: Solve the si.

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14 [Intermediate] • Target: Remainder and Factor Theorems

In the expansion of $(1+ax)^n$ in ascending powers of x, where n is a positive integer and a is a constant, the coefficient of x is -12 and the coefficient of x^2 is 60. Find the coefficient of x^3.

Final Answer: -160. Using the binomial theorem, the expansion of $(1+ax)^n$ is given by:
$(1+ax)^n = 1 + \binom{n}{1}(ax) + \binom{n}{2}(ax)^2 + \binom{n}{3}(ax)^3 + \ldots$ 1. **Equate the coefficient of x:**
$\binom{n}{1}a = na = -12 \Rightarrow a = -\frac{12}{n}$ 2. **Equate the coefficient of x^2:**
$\binom{n}{2}a^2 = \frac{n(n-1)}{2}a^2 = 60 \Rightarrow n(n-1)a^2 = 120$ Substitute $a = -\frac{12}{n}$: $n(n-1)\left(-\frac{12}{n}\right)^2 = 120$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

Final Answer: $-\frac{1}{2}$. By the Remainder Theorem, when a polynomial $f(x)$ is divided by a linear factor of the form $ax - b$, the remainder is $f\left(\frac{b}{a}\right)$. Here, the divisor is $2x - 1$, so we set $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$. Substitute $x = \frac{1}{2}$ into $f(x)$: $f\left(\frac{1}{2}\right) = 4\left(\frac{1}{2}\right)^3 - 6\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right) - 1 \ = 4\left(\frac{1}{8}\right) - 6\left(\frac{1}{4}\right) + \frac{3}{2} - 1$.

16 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

When the polynomial $\mathrm{P}(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x - 3$, the remainder is 10. What is the value of the constant a? Show your complete algebraic work.

Final Answer: -3. According to the Remainder Theorem, when a polynomial $\mathrm{P}(x)$ is divided by $x - c$, the remainder is equal to $\mathrm{P}(c)$. Here, the divisor is $x - 3$, so $c = 3$. We evaluate $\mathrm{P}(3)$:
$\mathrm{P}(3) = 2(3)^3 - 3(3)^2 + a(3) - 8$ $\mathrm{P}(3) = 2(27) - 3(9) + 3a - 8 = 54 - 27 + 3a - 8 = 19 + 3a$ Given that the remainder is 10: $19 + 3a = 10$ $3a = 10 - 19 = -9$ $a = -3$ Common misconceptions: - Choosing 3 results from a sign error when rearranging $19 + 3a = 10$ to.

17 [Challenging] • Target: Fractional Indices and Exponential Equations • Cross-Lesson: 6.01 Permutations and Combinations

The polynomial $p(x) = 2x^3 + ax^2 - 5x + 6$ is divisible by $x - 2$. Find the value of the constant a.

Final Answer: -3. By the Factor Theorem, if $p(x)$ is divisible by $x - 2$, then $p(2) = 0$. Substitute $x = 2$ into the polynomial expression: $p(2) = 2(2)^3 + a(2)^2 - 5(2) + 6$
$p(2) = 2(8) + 4a - 10 + 6 = 16 + 4a - 10 + 6 = 4a + 12$ Setting $p(2) = 0$: $4a + 12 = 0 \Rightarrow 4a = -12 \Rightarrow a = -3$
Common misconceptions: - Using the root $x = -2$ instead of $x = 2$ leads to
$2(-8) + 4a - 5(-2) + 6 = 4a = 0 \Rightarrow a = 0$. - A sign error when solving the linear equation $4a + 12 = 0$ incorrectly giv.

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18 [Challenging] • Target: Surds and Rationalising Denominators • Cross-Lesson: 7.01 Differentiation and Rates of Change

In the expansion of $(1+\frac{x}{4})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n? Show your complete algebraic work.

Final Answer: 17. Coeff of x is $\frac{n}{4}$. Coeff of x^2 is $\frac{n(n-1)}{2(16)}$. Given $\frac{n(n-1)}{2\cdot 16}=2(\frac{n}{4})\Rightarrow\frac{n-1}{2\cdot 4}=2\Rightarrow n-1=4(4)=16\Rightarrow n=17$.

19 [Challenging] • Target: Remainder and Factor Theorems • Cross-Lesson: 8.01 Integration and Kinematics

In the expansion of $(1+\frac{x}{5})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n? Show your complete algebraic work.

Final Answer: 21. Coeff of x is $\frac{n}{5}$. Coeff of x^2 is $\frac{n(n-1)}{2(25)}$. Given $\frac{n(n-1)}{2\cdot 25}=2(\frac{n}{5})\Rightarrow\frac{n-1}{2\cdot 5}=2\Rightarrow n-1=4(5)=20\Rightarrow n=21$.

20 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In the expansion of $(1+\frac{x}{6})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n? Show your complete algebraic work.

Final Answer: 25. Coeff of x is $\frac{n}{6}$. Coeff of x^2 is $\frac{n(n-1)}{2(36)}$. Given $\frac{n(n-1)}{2\cdot 36}=2(\frac{n}{6})\Rightarrow\frac{n-1}{2\cdot 6}=2\Rightarrow n-1=4(6)=24\Rightarrow n=25$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation \(|2x + 3| = 7 \).

a. \(x = 3 \) and \(x = -3 \)

b. \(x = -2 \) and \(x = -\frac{10}{2} \)

c. \(x = 2 \) only

d. **\(x = 2 \) and \(x = -\frac{10}{2} \)** ✓

Case 1: \(2x + 3 = 7 \implies 2x = 4 \implies x = 2 \). Case 2: \(2x + 3 = -7 \implies 2x = -10 \implies x = -\frac{10}{2} \).

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22 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|3x + 3| = 9$.

a. $x = 2$ and $x = -\frac{12}{3}$ ✓

b. $x = 3$ and $x = -3$

c. $x = -2$ and $x = -\frac{12}{3}$

d. $x = 2$ only

Case 1: $3x + 3 = 9 \implies 3x = 6 \implies x = 2$. Case 2: $3x + 3 = -9 \implies 3x = -12 \implies x = -\frac{12}{3}$.

23 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|4x + 3| = 11$.

a. $x = 2$ only

b. $x = 3$ and $x = -3$

c. $x = 2$ and $x = -\frac{14}{4}$ ✓

d. $x = -2$ and $x = -\frac{14}{4}$

Case 1: $4x + 3 = 11 \implies 4x = 8 \implies x = 2$. Case 2: $4x + 3 = -11 \implies 4x = -14 \implies x = -\frac{14}{4}$.

24 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|5x + 3| = 13$.

a. $x = 2$ and $x = -\frac{16}{5}$ ✓

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{16}{5}$

d. $x = 3$ and $x = -3$

Case 1: $5x + 3 = 13 \implies 5x = 10 \implies x = 2$. Case 2: $5x + 3 = -13 \implies 5x = -16 \implies x = -\frac{16}{5}$.

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25 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|6x + 3| = 15$.

a. $x = 3$ and $x = -3$

b. **$x = 2$ and $x = -\frac{18}{6}$ ✓**

c. $x = 2$ only

d. $x = -2$ and $x = -\frac{18}{6}$

Case 1: $6x + 3 = 15 \implies 6x = 12 \implies x = 2$. Case 2: $6x + 3 = -15 \implies 6x = -18 \implies x = -\frac{18}{6}$.

26 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|7x + 3| = 17$.

a. **$x = 2$ and $x = -\frac{20}{7}$ ✓**

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{20}{7}$

d. $x = 3$ and $x = -3$

Case 1: $7x + 3 = 17 \implies 7x = 14 \implies x = 2$. Case 2: $7x + 3 = -17 \implies 7x = -20 \implies x = -\frac{20}{7}$.

27 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|8x + 3| = 19$.

a. **$x = 2$ and $x = -\frac{22}{8}$ ✓**

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{22}{8}$

d. $x = 3$ and $x = -3$

Case 1: $8x + 3 = 19 \implies 8x = 16 \implies x = 2$. Case 2: $8x + 3 = -19 \implies 8x = -22 \implies x = -\frac{22}{8}$.

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28 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation \(|9x + 3| = 21 \).

a. \(x = 2 \) only

b. **\(x = 2 \) and \(x = -\frac{24}{9} \)** ✓

c. \(x = 3 \) and \(x = -3 \)

d. \(x = -2 \) and \(x = -\frac{24}{9} \)

Case 1: \(9x + 3 = 21 \implies 9x = 18 \implies x = 2 \). Case 2: \(9x + 3 = -21 \implies 9x = -24 \implies x = -\frac{24}{9} \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Fractional Indices and Exponential Equations

Evaluate the exact numerical value of $16^{-\frac{3}{4}} \times 27^{\frac{2}{3}}$.

a. $\frac{9}{8}$ ✓

b. $\frac{8}{9}$

c. $\frac{3}{4}$

d. $\frac{27}{16}$

$16^{-3/4} = (16^{1/4})^{-3} = 2^{-3} = 1/8$. $27^{2/3} = (27^{1/3})^2 = 3^2 = 9$. Product $= 9/8$.

30 [Intermediate] • Target: Fractional Indices and Exponential Equations

Solve the exponential equation $2^{2x+1} - 9(2^x) + 4 = 0$ for all real x.

a. $x = -1$ **and** $x = 2$ ✓

b. $x = 1$ and $x = 4$

c. $x = 0$ and $x = 2$

d. $x = -2$ and $x = 1$

Let $u = 2^x$: $2u^2 - 9u + 4 = 0 \Rightarrow (2u - 1)(u - 4) = 0 \Rightarrow u = 1/2 \Rightarrow x = -1$, or $u = 4 \Rightarrow x = 2$.

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31 [Intermediate] • Target: Fractional Indices and Exponential Equations

Simplify the index expression $\frac{(4x^3y^{-2})^{\frac{1}{2}}}{2x^{-1}y^3}$.

a. $x^{\frac{5}{2}}y^{-4}$ ✓

b. $2x^{\frac{5}{2}}y^{-4}$

c. x^2y^{-4}

d. $x^{\frac{5}{2}}y^2$

Numerator: $2x^{3/2}y^{-1}$. Divide by $2x^{-1}y^3$: $x^{3/2-(-1)}y^{-1-3} = x^{5/2}y^{-4}$.

32 [Intermediate] • Target: Surds and Rationalising Denominators

Rationalise the denominator and simplify $\frac{4+\sqrt{3}}{2-\sqrt{3}}$ into the form $a + b\sqrt{3}$.

a. $11 + 6\sqrt{3}$ ✓

b. $5 + 6\sqrt{3}$

c. $11 - 6\sqrt{3}$

d. $7 + 4\sqrt{3}$

Multiply top and bottom by $2 + \sqrt{3}$: $\frac{(4+\sqrt{3})(2+\sqrt{3})}{4-3} = 8 + 4\sqrt{3} + 2\sqrt{3} + 3 = 11 + 6\sqrt{3}$.

33 [Intermediate] • Target: Surds and Rationalising Denominators

Express $(3 - 2\sqrt{5})^2$ in the simplified form $p + q\sqrt{5}$.

a. $29 - 12\sqrt{5}$ ✓

b. $29 + 12\sqrt{5}$

c. $11 - 12\sqrt{5}$

d. $19 - 6\sqrt{5}$

$3^2 - 2(3)(2\sqrt{5}) + (2\sqrt{5})^2 = 9 - 12\sqrt{5} + 20 = 29 - 12\sqrt{5}$.

34 [Intermediate] • Target: Surds and Rationalising Denominators

Solve the radical equation $\sqrt{2x+7} - x = 2$ for all valid real solutions.

a. $x = 1$ **only** ✓

b. $x = 1$ and $x = -3$

c. $x = 3$ only

d. No real solutions

$\sqrt{2x+7} = x + 2 \Rightarrow 2x + 7 = x^2 + 4x + 4 \Rightarrow x^2 + 2x - 3 = 0 \Rightarrow (x-1)(x+3) = 0$. $x = 1$ works; $x = -3$ gives $\sqrt{1} - (-3) = 4 \neq 2$.

35 [Intermediate] • Target: Remainder and Factor Theorems

When $P(x) = 2x^3 + ax^2 - 5x + 6$ is divided by $(x - 2)$, the remainder is 16. Find the value of a.

a. $a = 1$ ✓

b. $a = -2$

c. $a = 3$

d. $a = 4$

$P(2) = 2(8) + a(4) - 5(2) + 6 = 16 + 4a - 10 + 6 = 4a + 12 = 16 \Rightarrow 4a = 4 \Rightarrow a = 1$.

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36 [Intermediate] • Target: Remainder and Factor Theorems

Given that $(2x - 1)$ is a factor of $f(x) = 2x^3 + 3x^2 - 8x + c$, find the constant c.

a. $c = 3$ ✓

b. $c = -3$

c. $c = 6$

d. $c = -2$

$f(1/2) = 2(1/8) + 3(1/4) - 8(1/2) + c = 1/4 + 3/4 - 4 + c = 1 - 4 + c = -3 + c = 0 \Rightarrow c = 3$.

37 [Intermediate] • Target: Remainder and Factor Theorems

Factorise completely the cubic polynomial $g(x) = x^3 - 4x^2 - 7x + 10$.

a. $(x - 1)(x + 2)(x - 5)$ ✓

b. $(x + 1)(x - 2)(x + 5)$

c. $(x - 1)(x - 2)(x - 5)$

d. $(x + 1)(x + 2)(x - 5)$

$g(1) = 1 - 4 - 7 + 10 = 0 \Rightarrow (x - 1)$ is a factor. Dividing gives $x^2 - 3x - 10 = (x + 2)(x - 5)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the coefficient of x^3 in the binomial expansion of $(1 + 2x)^6$.

a. 160 ✓

b. 80

c. 240

d. 120

Term is $\binom{6}{3}(1)^3(2x)^3 = 20(8x^3) = 160x^3$. Coefficient is 160.

39 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 8.01 Integration and Kinematics

Find the term independent of x (the constant term) in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

a. 240 ✓

b. 160

c. 60

d. 192

General term: $\binom{6}{r}(x^2)^{6-r}(2/x)^r = \binom{6}{r}2^r x^{12-3r}$. Constant when $12 - 3r = 0 \Rightarrow r = 4$. Term: $\binom{6}{4}2^4 = 15(16) = 240$.

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40 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In the expansion of $(1+ax)^5(1-x)$, the coefficient of x is 9. Find the value of a.

a. $a=2$ ✓

b. $a=1$

c. $a=3$

d. $a=-2$

$(1+5ax+\ \ldots)(1-x)=1+(5a-1)x+\ \ldots$. Given $5a-1=9\Rightarrow 5a=10\Rightarrow a=2$.

41 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In the expansion of $\left(1+\frac{x}{7}\right)^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n?

a. **29** ✓

b. 31

c. 28

d. 15

Coeff of x is $\frac{n}{7}$. Coeff of x^2 is $\frac{n(n-1)}{2(49)}$. Given $\frac{n(n-1)}{2\cdot 49}=2\left(\frac{n}{7}\right)\Rightarrow\frac{n-1}{2\cdot 7}=2\Rightarrow n-1=4(7)=28\Rightarrow n=29$.

42 [Challenging] • Target: Fractional Indices and Exponential Equations • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

In the expansion of $\left(1+\frac{x}{8}\right)^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n?

a. 17

b. **33** ✓

c. 35

d. 32

Coeff of x is $\frac{n}{8}$. Coeff of x^2 is $\frac{n(n-1)}{2(64)}$. Given $\frac{n(n-1)}{2\cdot 64}=2\left(\frac{n}{8}\right)\Rightarrow\frac{n-1}{2\cdot 8}=2\Rightarrow n-1=4(8)=32\Rightarrow n=33$.

43 [Challenging] • Target: Surds and Rationalising Denominators • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

In the expansion of $\left(1+\frac{x}{3}\right)^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n?

a. 12

b. 7

c. 15

d. **13** ✓

Coeff of x is $\frac{n}{3}$. Coeff of x^2 is $\frac{n(n-1)}{2(9)}$. Given $\frac{n(n-1)}{2\cdot 9}=2\left(\frac{n}{3}\right)\Rightarrow\frac{n-1}{2\cdot 3}=2\Rightarrow n-1=4(3)=12\Rightarrow n=13$.

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44 [Challenging] • Target: Remainder and Factor Theorems • Cross-Lesson: 6.01 Permutations and Combinations

In the expansion of $\left(1+\frac{x}{2}\right)^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n?

a. **9 ✓**

b. 11

c. 8

d. 5

Coeff of x is $\frac{n}{2}$. Coeff of x^2 is $\frac{n(n-1)}{2(4)}$. Given $\frac{n(n-1)}{2\cdot 4}=2\left(\frac{n}{2}\right)\Rightarrow\frac{n-1}{2\cdot 2}=2\Rightarrow n-1=4(2)=8\Rightarrow n=9$.

45 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 7.01 Differentiation and Rates of Change

The polynomial $P(x)=2x^2+ax-4$ is exactly divisible by $(x-6)$. What is the value of the constant a?

a. -14.33

b. 11.33

c. **-11.33 ✓**

d. -9.33

By the Factor Theorem, $P(6)=0\Rightarrow 2(6)^2+a(6)-4=0\Rightarrow 72+6a-4=0\Rightarrow a=-11.33$.

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